

Note- File assumed as Plant 2 is- ' Copy of PPT 25-26'

1. Overview

- Plant Type: Rooftop Solar
- Installed Capacity: 2340 kWp
- Data Period: Apr 2025 – Nov 2025

2. Key Assumptions

- GHI used as proxy for irradiance (since POA is not present)
- PR calculated using GHI is approximate PR used for relative comparison
- Threshold for downtime - Generation < 2000 kWh
- Loss values are estimated (not exact)

3. Key Performance Summary

Month	CUF	PR	Specific Yield	Downtime Days
Apr	12.9%	0.53	92.9	6
May	14.7%	0.74	109.5	1
Jun	14.7%	0.79	105.9	0
Jul	12.0%	0.80	89.4	0
Aug	11.0%	0.76	81.9	3
Sep	12.5%	0.80	89.8	0
Oct	10.7%	0.61	79.6	7
Nov	13.7%	0.71	98.5	1

4. Major Findings

Best Performance is observed in-

- **May & June**
 - CUF ~14.7%
 - PR ~0.75–0.80

- Stable operation

Worst Performance is observed in-

- **October**
 - CUF ~10.7%
 - PR ~0.61
 - Highest downtime (7 days)

Critical Fault Month

- **April**
 - PR dropped to ~0.53
 - Multiple downtime events
 - Inverter outage observed

5. Plant-Level Downtime Analysis

Summary

Month	Downtime Days	System	Weather	Loss (kWh)
Apr	6	5	1	~39,710
May	1	1	0	~7,469
Aug	3	1	2	~18,006
Oct	7	6	1	~39,036
Nov	1	1	0	~7,430

Key Insights

- System issues dominate downtime
 - April → 5/6 days system-related
 - October → 6/7 days system-related
- Weather impact is minor - only visible in August (monsoon)

Final Conclusion- Majority of energy loss is due to system-level downtime, not weather.

6. Inverter-Level Analysis (Vertical Issue)

Inverter 9 (April)

- Zero generation entire month
- Estimated loss: ~12–13 MWh

Inverter 8 (June)

- No data from 5th–17th June (~13 days)
- Estimated loss: ~6.36 MWh

Insight- Inverter-level faults contribute to losses but are smaller compared to plant-level downtime

7. Horizontal Issue (Day-wise Plant-Wide Events)

- April 1–7 → very low generation across all inverters
- High GHI (~6–7) but low generation

Interpretation: Indicates plant shutdown / grid outage, not weather issue

8. Loss Breakdown

Major Loss Contributors

- Plant Downtime Loss
 - April - ~39 MWh
 - October - ~39 MWh
- Inverter Loss
 - Inverter 9 - ~12–13 MWh
 - Inverter 8 - ~6.36 MWh
- Weather Loss
 - August (monsoon impact)

Key Insight- Plant downtime is the largest contributor to energy loss

9. PR & Degradation Analysis

- First-Year PR: ~0.75
- Current PR (best months): 0.75–0.80

Conclusion- No significant long-term degradation observed and performance drops are due to operational issues.

10. Correlation Insight (Strong correlation confirms healthy system behavior Low correlation indicates operational faults)

- June & September → ~0.98 (excellent)
- April → ~0.05 (fault condition)

11. Daily Performance Insight

- Average generation: ~7690 kWh
- High variability observed
- Very low generation days (<1000 kWh) indicate faults

12. Final Key Takeaways

- Plant performs well in high irradiance months
- Major losses due to system downtime (April & October)
- Inverter faults present but secondary
- Weather impact limited (mainly August)
- No major degradation observed

13. Recommendations

- Implement real-time downtime alerts
- Improve inverter-level monitoring
- Track downtime events with root cause logging